

Describe the absorption of light by red, green and blue cones in human eye. Draw the conceptual relationship between RGB and HIS models and describe the conversion of RGB to HIS and vice versa.

*absorption of light by red, green and blue cones in human eye
*relationship between RGB and HIS models
*conversion of RGB to HIS and vice versa

Describe the absorption of light by red, green and blue cones in human eye

The human eye has 6 to 7 million cones divided into three principal sensing categories, corresponding roughly to red, green, and blue.

65%: red

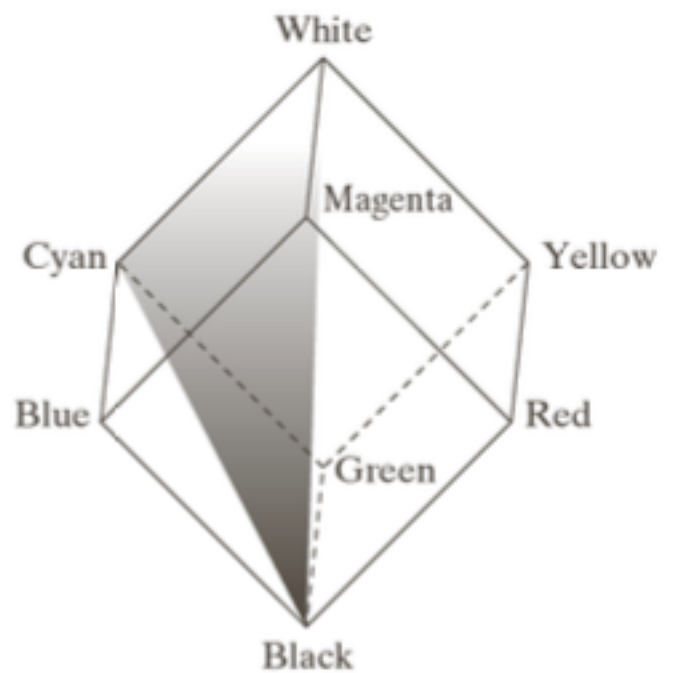
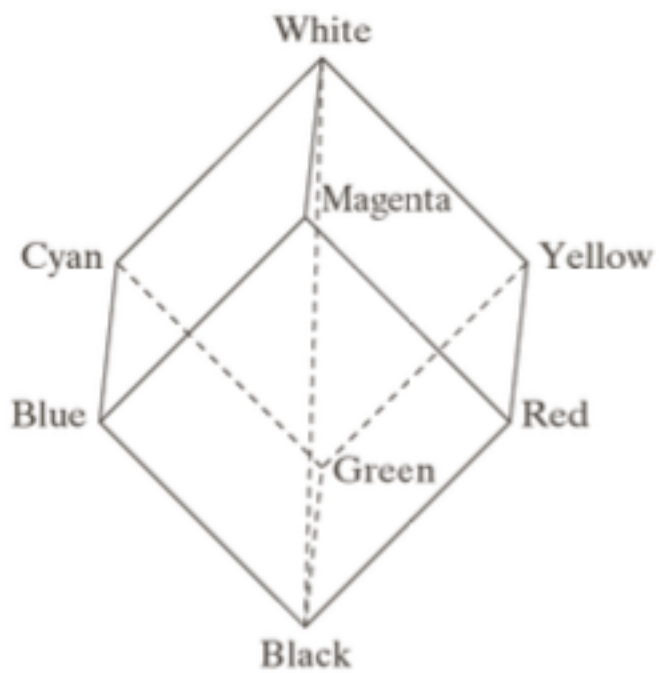
33%: green

2%: blue (blue cones are the most sensitive)

The characteristics generally used to distinguish one color from another are brightness, hue, and saturation.

- Brightness: the achromatic notion of intensity.
- Hue: dominant wavelength in a mixture of light waves, represents dominant color as perceived by an observer.
- Saturation: relative purity or the amount of white light mixed with its hue.

Draw the conceptual relationship between RGB and HIS models



Conversion of RGB to HIS and vice versa

Converting Colors from RGB to HIS:

Hue(H):

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

Where:

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$

Saturation (S):

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

Intensity (I):

$$I = \frac{1}{3} (R + G + B)$$

Converting Colors from HIS to RGB:

RG Sector ($0^\circ \leq H < 120^\circ$):

$$B = I(1 - S)$$

$$R = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

and

$$G = 3I - (R + B)$$

RG Sector ($120^\circ \leq H < 240^\circ$):

$$H = H - 120^\circ$$

$$R = I(1 - S)$$

$$G = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

and

$$B = 3I - (R + G)$$

RG Sector ($240^\circ \leq H < 360^\circ$):

$$H = H - 240^\circ$$

$$G = I(1 - S)$$

$$B = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

and

$$R = 3I - (G + B)$$